

Course Code: CS3005	Course Name: Theory of Automata
Instructor Name: Bakhtawar	
Student Roll No:	Section No:

Instructions:

- Read each question completely before answering it. There are **5 questions** on **2 pages**.
- In case of any ambiguity, you may make assumption. But your assumption should not contradict any statement in the question paper.
- **Cutting and over-writing is strictly discouraged.**

Total Time: 30 minutes**Maximum Points:** 35**Question # 1(5+5+5 Points)**

Write down the RE for following languages.

- Language of the words that start with bb and end aa with and have baa as a substring.
- Define language such that it should contain exactly one occurrence of double letter over alphabetical set {a,b}
- $L = \{a^n b^m \mid (n+m) \text{ is even}\}$

Question # 4 (5+5+5 Points)

Design the FA for following languages

- $L = \{aba, bab, bbbb, aaa\}$
- $L = \{vwz \mid |v| = 4, v \in \{a, b\}^*, z \in \{a, b\}^+, w \in \{a, b\}^*\}$
- Design a DFA such that the string does not contain '00' as a substring.

Question # 5 (5 Points)

Convert the following NFA to DFA.

